

Multilingual Characterization of Entities from Online News

Carolina Gonzalez

University of California, Berkeley
Carolina.Gonzalez@Berkeley.edu

Anshul Zutshi

University of California, Berkeley
anshulzutshi@ischool.berkeley.edu

Abstract

Advancements in natural language processing (NLP) have significantly improved the ability to classify entities across diverse texts, with hierarchical classification emerging as a crucial technique for capturing entity roles. Entity classification systems are increasingly applied in multilingual contexts, where challenges such as lexical variation and cultural differences require broad and adaptable models. This study explores hierarchical entity classification using transformer models, emphasizing their effectiveness in multilingual settings. This study focuses on the two-tier classification of entities in online news in four different languages. The first level of the hierarchy are main roles and the secondary level are fine-grain roles which are more detailed classifications of the main roles. The articles analyzed and task annotations are as per SEMEVAL 2025 Task 10 "Multilingual characterization and extraction of narratives from online news" Task 1. (1) This paper compares the two approaches for hierarchical classification using Roberta (5) and Llama 3.2 for four languages: English, Portuguese, Bulgarian, and Hindi.

1 Introduction

Entity classification has evolved significantly over the years, becoming a critical component of natural language processing (NLP) with broad applications in information extraction, knowledge graph construction, and narrative analysis. Early systems relied heavily on rule-based approaches, leveraging predefined lexicons and ontologies to identify and classify entities. While effective within narrow domains, these methods struggled to adapt to new contexts or datasets (10). Around 2007, the introduction of machine learning models marked a shift toward feature-based approaches. These models, although more flexible, often required extensive manual feature engineering and remained domain-specific, limiting their generalizability to unseen

data (11).

The advent of deep learning further revolutionized entity classification by introducing models that could learn hierarchical representations of text. Recurrent Neural Networks (RNNs), including their advanced forms like LSTMs and GRUs, began incorporating character-level, subword-level, and word-level embeddings. These models proved more robust when applied to new domains and datasets, significantly reducing the dependency on explicit feature extraction. However, their reliance on large volumes of annotated training data introduced a new challenge, as the manual labeling process is both time-consuming and resource-intensive, often creating a bottleneck for real-world applications (12).

The current era, defined by transformer architectures such as BERT (13), RoBERTa (14), and Llama (15), has addressed many of these limitations. These models, pretrained on massive multilingual corpora, excel at capturing contextual relationships and adapting to diverse tasks with minimal task-specific tuning. Hierarchical entity classification, in particular, has benefited from their ability to model both coarse-grained and fine-grained role distinctions.

2 Background

The SemEval 2025 Task 10, Multilingual Characterization and Extraction of Narratives from Online News, focuses on identifying and analyzing hierarchical roles in narratives related to the Ukraine-Russia War and Climate Change. The task consists of two sub-tasks: Sub-task 1, which involves broad role classifications (e.g., Antagonist, Protagonist, Innocent), and Sub-task 2, which refines these roles into more specific distinctions (e.g., Foreign Adversary, Spy) (1). These narratives are inherently multilingual, spanning Bulgarian, English, Hindi, and Portuguese datasets, and present challenges

due to linguistic diversity, cultural context, and label imbalances.

Modern transformer-based architectures such as RoBERTa and Llama are well-suited to these challenges. RoBERTa, with its robust pretraining optimization (14), has demonstrated strong performance in multilingual settings. Similarly, Llama leverages advancements in efficient attention mechanisms and parameter tuning to capture long-range dependencies in text, making it particularly effective for hierarchical classification tasks (15). These architectures allow for the fine-grained representation of entities, critical for identifying nuanced propaganda techniques within narratives.

Several studies have informed the design of this work. Research on multilingual representations, such as "Multilingual Name Entity Recognition and Intent Classification employing Deep Learning Architectures" (2), emphasizes the importance of shared semantic representations for cross-lingual generalization. Furthermore, Koreeda's exploration of cross-lingual multi-task strategies demonstrates the value of leveraging shared representations across tasks and languages (4).

The dataset's challenges are further compounded by the hierarchical nature of the labels and the need for balance across languages. Data augmentation strategies, inspired by works such as Zhang's "mGTE: Generalized Long-Context Text Representation and Reranking Models for Multilingual Text Retrieval" (6), offer innovative methods to address these issues. These strategies enable the enrichment of underrepresented labels, improving model robustness and generalization performance.

This research integrates these insights to develop robust multilingual classification systems capable of navigating the complexities of diverse datasets and cultural contexts. By refining transformer models and tailoring them to hierarchical entity classification, this work advances the field of multilingual NLP and sets a benchmark for tackling real-world narrative detection challenges.

3 Methods

Data preprocessing involved tokenizing each news article using language-specific tokenization tools compatible with the Roberta and Llama 3 architectures.

3.1 Data Acquisition and Preprocessing

We obtained the data from the ACL conference website (1) and loaded the data into google drive and then utilized google colab to build, load, and run models on the imported data.

3.2 RoBERTa Model

Our baseline model leverages Roberta XLM large, a transformer-based model optimized for language understanding tasks. Given its architecture and training data with more than 100 languages, RoBERTa is well-suited to multilingual tasks, allowing us to fine-tune it for entity classification across the four languages specified.

Sub-task 1 required the RoBERTa model to identify whether an entity was an Innocent, Protagonist, or Antagonist. These roles were further specialized in sub-task 2 which broke down each role into 7-8 options ranging from Spy to Martyr.

The threshold for prediction for all of the 22 class categories for sub-task 2 (fine grain roles) was optimized on f1-score for all the experiments. The optimization relied on the the training set and assigned a threshold to each class. The fine-tuned classes with a score higher than their threshold were chosen as classes for their corresponding entity.

3.3 Experiment 1: Standard fine-tuning for Baseline RoBERTa Model

We first experimented with combining the losses of the two models each trained for their separate sub-tasks, and attempting to minimize this loss based on the training data to create the best model. The model included one loss for Sub-task 1 and one loss for Sub-task 2. The total loss for the training steps is the sum of the losses for the two tasks with equal weight, lambda equal to 1. This led to us having very high performance on the first sub-task, however, we had very lackluster results on the second sub-task.

3.4 Experiment 2: RoBERTa Model with Lambda 0.5

As a result, we then attempted to alter the weights assigned to each sub-task when combining the losses together. We decided to use a lower weight (0.5) for the main role classification and better weight (1) for the fine grain role task, in order to improve model performance for the fine grain classification task. This led to a marginal increase in performance for the fine grained roles with a

marginal decrease for the main roles which was to be expected.

3.5 Experiment 3: Using a Combined Vector for RoBERTa Model

Since combining the losses with different weights did not seem to work as intended, we decided to create a combined prediction vector that contains predicted labels for both the main roles and fine-grained roles. The first part of the vector will include the one-hot encoded prediction of the entity’s main role while the second part of the vector will include the predicted values of the secondary roles. This allows for the model to train on both the main role and fine-grained role simultaneously. Thus we can then calculate the singular loss for the training step using PyTorch BCEWithLogitsLoss. The prediction was determined by applying a sigmoid function to the logits, where values exceeding an optimal threshold were classified as positive predictions. This led to a slight overall increase for both main and fine-grained roles.

3.6 Experiment 4: Data Augmentation on Roberta

The main issue in increasing the performance of the classification models was related to having very few examples labeled for some of the fine-grained roles in different languages. Therefore, we experimented with data augmentation to help increase the data volume to feed into our model.

We experimented with targeted data augmentation using the library NLPAUG Contextual-WordEmbsAug on only labels with low data volume versus applying data augmentation to the entire dataset. We determined underrepresented labels by calculating the median number of labels for a role and selecting all roles that fell below that mark as underrepresented labels. These labels ended up being Corrupt, Rebel, Saboteur, Exploited, Underdog, Traitor, Forgotten, Scapegoat, Bigot, Martyr, and Spy.

We also tested both data insertion (inserting additional words into an article), data substitution (replacing a handful of words with synonyms), and them in combination to determine the best way to augment our dataset to produce better results.

3.7 Experiment 5: Llama 3

We also experimented with a different type of model, Llama, in the form of Llama 3.2 1B instruct, Llama 3.2 3B instruct, and Llama 3.1 8B

instruct. After reading more literature on the Llama models through Huggingface, we found that Llama 3.1 8B was recommended for entity classification tasks due to its superior fine-tuning capabilities and ability to handle multilingual datasets effectively, though its decoder-only architecture requires careful adaptation for classification compared to encoder-based models like RoBERTa. (8)

We experimented with a number of different prompts, ranging from explicit instruction, including the taxonomy of the roles to a general instruction to classify the entities among the 22 main and fine-grained roles. We found that the simpler prompts actually led to marginally better results, but further research likely should be done to identify the optimal prompt.

Due to it being a text generation model, evaluating the Llama’s model performance was significantly more difficult as we had to isolate the labels predicted amidst the high volume of text generated. As a result, the performance of the Llama model family was much worse than that of the RoBERTa-type model despite fine-tuning efforts.

Due to the small dataset, the Llama model, with its extensive 8 billion parameters, demonstrated clear signs of overfitting to the training data. This was evident as the training loss continued to decrease steadily while the validation loss plateaued, failing to reflect improvements in generalization. To address this issue, we implemented several strategies aimed at mitigating overfitting.

First, we introduced dropout to the LoRA (Low-Rank Adaptation) layers, which helped regularize the model by randomly deactivating a fraction of neurons during training. (9) Second, we reduced the learning rate to slow the rate of parameter updates, allowing the model to converge more effectively without over-adjusting to the nuances of the training data. Finally, we dramatically reduced the number of training epochs from 18 to 5 to prevent the model from learning spurious correlations in the limited dataset. These adjustments collectively aimed to balance the model’s capacity with the constraints of the available data, improving its performance on unseen validation samples.

4 Results and Discussion

The performance for the models was compared for all 4 languages and roles were measured using a micro f1-score that factors in class imbalances when determining the model with the best performance.

The baseline for this experiment was the RoBERTa XLM Large model, measuring its ROC AUC score on the three main categories: protagonist, antagonist, and innocent (5). We used ROC AUC score because it illustrates the trade-off between the True Positive Rate (TPR) and False Positive Rate (FPR), offering a more holistic view of the model’s performance compared to other metrics such as accuracy or precision. The results highlight the variability in performance among the 4 languages.

	English	Hindi	Portuguese	Bulgarian	Average
Protagonist	0.702	0.776	0.905	0.675	0.820
Antagonist	0.787	0.804	0.844	0.763	0.818
Innocent	0.722	0.853	0.823	0.843	0.828
Average	0.737	0.811	0.857	0.760	

Figure 1: ROC-AUC Scores of the main roles for the Baseline RoBERTa model (Experiment 1)

	English	Hindi	Portuguese	Bulgarian	Average
Protagonist	0.63	0.769	0.953	0.709	0.824
Antagonist	0.653	0.766	0.811	0.815	0.798
Innocent	0.662	0.865	0.779	0.845	0.818
Average	0.648	0.800	0.848	0.790	

Figure 2: ROC-AUC Scores of the main roles for the RoBERTa model with a combined vector (Experiment 3)

	English	Hindi	Portuguese	Bulgarian	Average
Protagonist	0.665	0.761	0.956	0.695	0.829
Antagonist	0.782	0.743	0.853	0.795	0.812
Innocent	0.731	0.877	0.832	0.834	0.846
Average	0.726	0.794	0.880	0.775	

Figure 3: ROC-AUC Scores of the main roles for the best data augmented RoBERTa model (Experiment 4)

	Llama 3.1 8B	Llama 3.1 8B finetuned	Llama 3.2 1B	Llama 3.2 1B finetuned	Llama 3.2 3B	Llama 3.2 3B finetuned	Best RoBERTa model
Protagonist	0.37	0.24	0.15	0.33	0	0	0.65
Antagonist	0.67	0.6	0.5	0.27	0.20	0.53	0.73
Innocent	0.40	0.37	0.21	0.1	0	0	0.59
Average	0.480	0.403	0.287	0.233	0.067	0.177	0.66

Figure 4: Micro-f1 scores of the main roles for the Llama models pre and post fine-tuning(Experiment 5)

	English	Hindi	Portuguese	Bulgarian	Average
Protagonist	0.08	0.51	0.12	0.17	0.34
Antagonist	0.73	0.61	0.67	0.68	0.66
Innocent	0.25	0.38	0.53	0.17	0.39
Average	0.353	0.50	0.44	0.34	

Figure 5: Micro-f1 scores of the main roles for the Llama models broken down by language (Experiment 5)

We observe in Figure 1 that the baseline RoBERTa model achieves strong performance

across all languages and roles, with the highest scores recorded on the Portuguese dataset and the Innocent main role, yielding average ROC AUC scores of 0.8572 and 0.82758, respectively. The model’s consistent average AUC-ROC score exceeding 0.8 across approximately 500 cumulative datapoints suggests that its multilingual training and fine-tuning are effective at capturing the general structure of the data.

Figure 2 shows the combined vector RoBERTa model performing comparably to the baseline in overall role classification. However, the model exhibits notable variation across languages, performing worse in English but significantly better in Bulgarian. The standout performance in Portuguese for protagonist roles, with a ROC AUC score of 0.953, might be due to the model’s ability to align the hierarchical structure of roles more effectively for this language.

In Figure 3, the results from the best-performing data augmentation RoBERTa model reveal improvements across Portuguese and all main roles: Innocent, Antagonist, and Protagonist. The superior performance is likely attributable to the increased data diversity and representation achieved through augmentation techniques. By addressing the class imbalance in fine-grained roles, the model appears to generalize better, benefiting the accuracy of main roles due to the hierarchical relationship between main and fine-grained roles. This highlights the importance of data volume and distribution in optimizing multilingual classification tasks.

Because the Llama models we use generate text rather than probabilities for labels, the ROC-AUC score is not an appropriate metric to evaluate performance for the Llama model. Instead we should use the micro f1 score to balance the effect of precision and recall and account for class imbalances to evaluate our Llama models and compare their performance against the best performing RoBERTa model that utilizes both data insertion and substitution. When examining the scores in Figure 4, we can see that all Llama models perform much worse than our best performing RoBERTa model. In particular, we can see that all of the Llama models struggle in correctly classifying both Innocent and Antagonist roles, and our best performing model Llama 3.1 8B without finetuning performs significantly worse than our best RoBERTa model in all main roles. This is likely due to a number of factors: the complex architecture of Llama not adapting to the dataset well, the difficulty in entity identifica-

	All languages	English	Bulgarian	Hindi	Portuguese
RoBERTa xlm					
Million of Training Tokens		55608	5487	1715	8405
Main Roles					
Antagonist	1001	220	228	291	262
Innocent	447	38	79	179	151
Protagonist	580	81	42	274	183
Fine Grain Roles					
Bigot	18	5	3	8	2
Conspirator	95	30	39	11	15
Corrupt	78	13	22	15	28
Deceiver	97	23	9	39	26
Exploited	40	5	9	19	7
Foreign Adversary	294	31	65	106	92
Forgotten	21	1	4	11	5
Guardian	221	29	3	107	82
Incompetent	116	29	52	20	15
Instigator	159	41	40	39	39
Martyr	17	9	5	1	2
Peacemaker	132	14	16	65	37
Rebel	47	8	11	14	14
Saboteur	45	17	2	13	13
Scapegoat	20	6	0	9	5
Spy	14	2	0	12	0
Terrorist	78	14	17	16	31
Traitor	24	7	6	7	4
Tyrant	102	24	31	27	20
Underdog	37	12	2	21	2
Victim	378	27	67	148	136
Virtuous	171	13	7	101	50
Total Examples	2028	339	349	744	596

Figure 6: Training data language split

tion amongst the verbose text generate of the Llama model, and struggling to fine-tune due to the limited dataset.

When looking at the Llama 3.1 8B model’s performance across the 4 different languages in Figure 5, we can glean a number of things. Firstly, we can see that model excels in classifying antagonist labels (0.66) when compared to its performance in protagonist and innocent, 0.34 and 0.39, respectively. This disparity suggests that the model may capture certain features unique to antagonist roles more effectively. However, when comparing its performance across languages, interestingly it only performs well classifying the protagonist class in Hindi at 0.51, triple the closest score in any of the other 3 languages. This may indicate a language-specific alignment or data distribution bias in Hindi that facilitates improved classification for this role. Overall, the Llama model’s performance remains significantly below the best RoBERTa model and fails to surpass even the baseline RoBERTa, suggesting that its architecture or training may be less suited to this classification task.

Re-examining the RoBERTa model’s performance across languages reveals a general ranking from best to worst: Portuguese, Hindi, Bulgarian, and English. Interestingly, as shown in Figure 5, English has the largest number of training tokens but the fewest total labels, whereas Hindi has the

fewest tokens but the highest label count, closely followed by Portuguese. This contrast suggests that label volume plays a more critical role in model performance than the sheer number of training tokens. Improving the model’s performance in English may thus require increasing the number of labeled training samples, particularly for fine-grained role distinctions.

The performance differences between languages are likely influenced by lexical variations, as the four languages have distinct scripts and roots—English with Germanic roots, Portuguese originates from Latin, Hindi is derived from Sanskrit, while Bulgarian is a Slavic language. Figure 5 also highlights that certain roles, such as Spy, have no examples in Bulgarian and Portuguese, while Martyr has only a handful of instances across all languages. Consequently, data augmentation is likely needed to bolster the volume of low-frequency, fine-grained roles for better model performance.

This transitions us into our discussion on the second sub-task, fine grained roles. We first wanted to evaluate our performance using both RoBERTa and Llama 3.2 models and compare the results on the dataset. However, when conducting our analysis using RoBERTa, we quickly discovered that since there were 22 fine grained roles, and not only was there a great similarity between the roles but each role had a very limited dataset in all languages. In fact, 8 of the 22 fine-grained roles had fewer than 10 articles in multiple languages: Bigot, Exploited, Forgotten, Martyr, Scapegoat, Spy, Traitor, and Underdog.

This led to our performance on the fine-grained roles to be poor, with an average f1-micro score of 0.26 across all roles. We chose to use f1-micro score as an evaluation metric as it balances precision and recall while also taking into account major class imbalances within the fine-grained roles. As a result of the poor performance of the baseline model and low data volume, we decided to implement data augmentation through the package nlpaug in an attempt to generate additional articles for each fine-grained role. There were two methods of data augmentation we attempted: insertion and substitution.

The first method, data insertion, involves adding words to an article while maintaining its original meaning. This helps diversify the text corpus, which is crucial for improving model performance, particularly when the dataset is limited. Our first ex-

	Baseline RoBERTa model w/ Loss Combined	RoBERTa model w/ Combined Vector	RoBERTa model w/ data insertion (3x)	RoBERTa model w/ data insertion & substitution	RoBERTa model w/ data insertion & substitution for underrepre- sented labels
Bigot	0.02	0.03	0	0.02	0
Conspirator	0.09	0.49	0.37	0.45	0.46
Corrupt	0.07	0.06	0.09	0.1	0
Deceiver	0.09	0.05	0.16	0.11	0.13
Exploited	0.05	0.03	0	0	0
Foreign Adversary	0.3	0.38	0.47	0.4	0.46
Forgotten	0.03	0	0	0.02	0
Guardian	0.38	0.43	0.34	0.45	0.46
Incompetent	0.11	0.14	0.22	0.21	0.18
Instigator	0.16	0.27	0.3	0.29	0.23
Martyr	0.07	0	0	0.01	0
Peacemaker	0.2	0.25	0.22	0.2	0.29
Rebel	0.1	0.04	0	0.08	0
Saboteur	0.05	0.03	0.15	0.1	0
Scapegoat	0.06	0.02	0.2	0.17	0
Spy	0.01	0.01	0.4	0	0
Terrorist	0.08	0.04	0.23	0.09	0.12
Traitor	0.02	0.06	0	0.03	0
Tyrant	0.11	0.12	0.2	0.14	0.12
Underdog	0.11	0.05	0	0	0
Victim	0.73	0.59	0.46	0.56	0.55
Virtuous	0.27	0.39	0.25	0.32	0.32
F1-Micro Average	0.22	0.26	0.45	0.32	0.48

Figure 7: Comparing F1-Micro Scores of the fine grained roles for the RoBERTa models with and without data augmentation

periment was to simply data insert with anywhere from 1-10 words into each excerpt which double our data pool and drastically improved our results. We then hypothesized if we vary the number of words added and insert more than once, we can triple or even quadruple our dataset which could lead to further improvements. We inserted 1-3, 4-6, and 7-10 words per article, quadrupling the data volume. This resulted in a significant increase in the F1-micro score, from 0.26 to 0.45—approximately a 73 percent improvement versus the baseline model and a marginal improvement when comparing to the first data augmented model.

The second method, data substitution, replaces a set number of words in the article with synonyms. We experimented with using substitution in combination with insertion 1-3 words with synonyms—was combined with the insertion of 1-3 words per article. This tripled the data volume and increased the F1-micro score to 0.32, surpassing the baseline model’s performance. This suggests that data augmentation not only enhances the model’s exposure to different phrasing but also helps balance the learning across various roles.

Finally, focusing on underrepresented labels through data augmentation led to the highest F1-micro score of 0.48. This improvement highlights the importance of addressing class imbalance, as the model performed best when it had more balanced representation of the fine-grained roles.

F1-score	All Languages	English	Bulgarian	Hindi	Portuguese
Antagonist	0.73	0.8	0.79	0.62	0.71
Innocent	0.6	0.38	0.65	0.6	0.59
Protagonist	0.64	0.44	0.21	0.63	0.82
Bigot	0.03	0.08	0.08	0	0
Conspirator	0.49	0.36	0.59	0	0.22
Corrupt	0.06	0	0.1	0	0
Deceiver	0.05	0	0	0.18	0.09
Exploited	0.03	0	0.08	0.02	0
Foreign Adversary	0.38	0	0.46	0.33	0.45
Forgotten	0	0	0	0	0
Guardian	0.43	0.42	0	0.43	0.47
Incompetent	0.14	0	0.23	0	0
Instigator	0.27	0.19	0.39	0.21	0.25
Martyr	0	0	0	0	0
Peacemaker	0.25	0	0	0.25	0.29
Rebel	0.04	0.09	0.05	0.04	0
Saboteur	0.03	0.08	0	0.03	0.04
Scapegoat	0.02	0	0.08	0.01	0
Spy	0.01	0.03	0	0.01	0
Terrorist	0.04	0	0.14	0	0
Traitor	0.06	0	0.09	0	0
Tyrant	0.12	0.12	0.11	0.16	0.08
Underdog	0.05	0	0.06	0.08	0
Victim	0.59	0.5	0.61	0.55	0.63
Virtuous	0.39	0	0	0.45	0.33
micro	0.26	0.24	0.27	0.24	0.27
macro	0.22	0.14	0.19	0.18	0.2
weighted	0.49	0.4	0.51	0.46	0.54
samples	0.26	0.26	0.27	0.24	0.29

Figure 8: Experiment 3 results

Thus, the data augmentation strategies proved particularly effective in improving overall model performance by increasing data diversity and addressing underrepresented labels.

5 Next Steps

One way to further improve our results would be to obtain additional labeled data for our model to train on. Despite the fact that data augmentation helped improved the model, there is a limit to how much the model can improve only solely with duplicating data and slightly altering it. Furthermore, there are some roles in certain languages that contain 0 examples in the training data like Scapegoat and Spy in Bulgarian and Spy in Portuguese which makes it impossible for any model to correctly classify these roles.

In addition to gathering more labeled data, perhaps there are more things we could have done to obtain better results using a Llama model. Perhaps some further research into prompt engineering is warranted and can be used to obtain better results for Llama and perhaps discover a method to build a Llama model that can outscore the best RoBERTa model.

Another potential way to improve the RoBERTa models could be to refine the thresholds for fine-tuned roles to be chosen to improve model performance on sub-task 2. This threshold was based on the training data for fine-tuned roles of which we have seen have a limited sample for certain roles

in different languages. As a result, perhaps there should be some regularization term to adjust for this low volume to reduce the variability of the thresholds across the fine-tuned roles.

There are also other model alternatives to RoBERTa XLarge like LUKE, which incorporates knowledge graph embeddings, making it adept at political entities with rich relationships that could be tested as well to see if its performance is comparable to RoBERTa (7).

6 Conclusion

In this study, we evaluated the performance of RoBERTa and Llama models for entity classification tasks, specifically focusing on the role characterization of entities in multilingual news data. Our experiments demonstrate that the RoBERTa XLM Large model is a strong baseline for multilingual entity classification, consistently achieving high performance across languages, especially in Portuguese. However, we found that class imbalance, especially in fine-grained roles, limited the model's performance on less-represented categories.

We explored several methods to improve performance, including adjusting the loss function calculations through weights and vector combining, fine-tuning and hyperparameter tuning, and using data augmentation techniques to increase the data corpus. Data augmentation, in particular, proved to be a powerful tool in improving classification accuracy, particularly for underrepresented fine-grained roles. The RoBERTa model using both data insertion and substitution showed the most significant improvements, especially for roles with minimal training data.

Despite the strengths of RoBERTa, our results showed that the Llama models, designed for text generation tasks, performed suboptimally in entity classification. The large parameter size coupled with a limited, specialized dataset leads us to conclude that Llama's architecture is not ideally suited for this particular task, at least when applied under the settings we tested.

We also found a wide range of performance across languages, with models performing the best overall on Portuguese articles yet, shocking, struggled with English articles. The similarity between many of the fine-tuned roles such as Rebel and Underdog makes the challenge of correctly classifying these entities extremely difficult, even if a human were to attempt it. However, RoBERTa does a re-

markable job in classifying entities into their three main roles: Innocent, Antagonist, and Protagonist while Llama's high number of parameters end up becoming a detriment in the face of a low data sample.

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